

Lab 02 - Composite behaviors

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This lab exercise requires that the robot goes in the direction of the light while avoiding obstacles.

Idea

My idea is to combine the two main behaviors (phototaxis and collision avoidance) through continuous behavior blending. Each behavior produces a steering contribution, and the final wheel velocities are obtained by combining these contributions into a single continuous motor command.

Controller architecture - Possible solution

The controller is implemented in `composite_behaviors.lua` and includes the following components:

- **Sensors:** arrays of light sensors (`robot.light`) and proximity sensors (`robot.proximity`).
- **Phototaxis:** computes the normalized difference between right and left light sides (`light_diff`) and produces a steering command proportional to `LIGHT_GAIN * light_diff * MAX_VELOCITY`.
- **Collision avoidance:** sums proximity values on the left and right sides to compute `prox_diff` and calculates `obstacle_turn = OBSTACLE_GAIN * prox_diff * MAX_VELOCITY`.
- **Speed reduction:** the base speed is scaled by `speed_factor = 1 - (max_prox * SPEED_REDUCTION)` to slow down near obstacles.
- **Gap detection:** if the average of front sensors is low while side sensors report high values, the controller increases `speed_factor` and attenuates steering to safely traverse a narrow passage. The controller uses a threshold-free continuous measure of gapness computed from front and side proximity readings. A `gap_weight` (0...1) is derived from the relative difference between side and front signals and smoothly increases `speed_factor` while attenuating steering commands. This avoids binary decisions and produces graceful behavior when entering narrow passages.
- **Random walk:** when neither light nor obstacles are significantly perceived, the controller injects a small random perturbation into the steering signal to promote exploration and avoid deadlocks.
- **Continuous steering:** instead of using discrete left/right turning states, the controller computes wheel velocities continuously as `left_v = base_speed + total_turn` and `right_v = base_speed - total_turn`, where `total_turn` is obtained by combining phototaxis and obstacle avoidance steering contributions. This produces smoother trajectories and reduces oscillatory behavior.

Expected behavior

- With a dominant light source and open space, the robot converges towards the light.
- When approaching an obstacle the robot steers away using `obstacle_turn` and reduces speed via `speed_factor`.
- In narrow passages (gaps) the controller detects a clear center with close sides, reduces steering and temporarily increases relative speed to cross the gap smoothly.
- When no significant objects are present and no light is visible, the robot performs a simple exploratory random walk to avoid getting stuck.